

MIP_EXECUTION_ROADMAP.md

Mainframe Intelligence Platform

Execution Roadmap

Version 1.0

Strategic Objective

Build a reusable enterprise platform that reduces modernization risk and accelerates understanding of legacy systems.

Success Metrics

Reduce:

- Dependency on SMEs
- Onboarding time
- Impact analysis effort
- Modernization risk

Increase:

- Visibility
 - Productivity
 - Knowledge retention
 - Change confidence
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Phase 0

Vision & Foundation

Duration:

2 Weeks

Deliverables:

- Vision Document
- Architecture Document

- Metadata Model
- Knowledge Graph Model
- Governance Model

Success Criteria:

Leadership alignment.

Phase 1

Repository Discovery

Duration:

4 Weeks

Objective:

Understand what exists.

Capabilities:

- Repository scanning
- File classification
- Inventory creation

Deliverables:

- Artifact Catalog
- Inventory Dashboard

Questions Answered:

- What exists?
 - Where is it located?
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Phase 2

Metadata Extraction

Duration:

6 Weeks

Capabilities:

COBOL

Extract:

- PROGRAM-ID
- CALL
- COPY
- EXEC SQL

JCL

Extract:

- JOB
- PROC
- EXEC PGM

DB2

Extract:

- Tables
- Views

Deliverables:

- Program Catalog
- Job Catalog
- Table Catalog

Questions Answered:

- What programs exist?
 - What jobs exist?
 - What tables exist?
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Phase 3

Knowledge Graph

Duration:

6 Weeks

Capabilities:

- Relationship discovery
- Dependency graph

Deliverables:

- Graph Model
- Relationship APIs

Questions Answered:

- Who calls whom?
 - What executes what?
 - What updates what?
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Phase 4

Dependency & Impact Analysis

Duration:

6 Weeks

Capabilities:

- Root discovery
- Dependency tracing
- Impact analysis

Questions Answered:

- What breaks if changed?
 - What depends on this?
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Phase 5

Business Understanding

Duration:

8 Weeks

Capabilities:

- Business rule extraction
- Capability mapping

Deliverables:

- Business Capability Catalog

- Rule Repository

Questions Answered:

- Why does this exist?
 - What business function does it support?
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Phase 6

AI Copilot

Duration:

8 Weeks

Capabilities:

Natural language questions.

Examples:

- Show payment flow.
- Explain program.
- Trace batch execution.
- Show impacted systems.

Deliverables:

- Enterprise Copilot
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Phase 7

Modernization Intelligence

Duration:

10 Weeks

Capabilities:

- Service identification
- Domain discovery
- API candidate generation
- Event candidate generation

Questions Answered:

- What should become a service?
 - What can be decomposed?
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Phase 8

Transformation Assistance

Duration:

12 Weeks

Capabilities:

- COBOL to Java guidance
- Test generation
- Migration recommendations

Deliverables:

- Modernization Assistant
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Team Structure

Phase 1-3

Small Team:

- 1 Architect
 - 2 Engineers
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Phase 4-6

Medium Team:

- 1 Architect
 - 4 Engineers
 - 1 Data Engineer
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Phase 7-8

Expanded Team:

- Architects
 - Platform Engineers
 - AI Engineers
 - Modernization Specialists
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Risks

Technical

- Parsing complexity
 - Legacy variations
 - Missing metadata
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Organizational

- SME availability
 - Repository access
 - Governance delays
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Non-Goals

Not a replacement for:

- Source control
- Mainframe runtime
- Deployment tooling

Initial focus:

Understanding before transformation.

MVP Definition

MVP is complete when users can ask:

1. Which jobs execute program X?
2. Which programs call program X?
3. Which copybooks does program X use?
4. Which tables does program X update?
5. Show end-to-end job flow.
6. Identify root programs.

If these six capabilities work reliably, MIP has achieved foundational value.

Long-Term Vision

MIP becomes the enterprise operating system for legacy application intelligence and modernization.

Every modernization effort starts with MIP.

Every impact analysis uses MIP.

Every onboarding engineer learns through MIP.

Every transformation initiative is guided by MIP.